

Photometric Analysis and Age Determination of the Globular Cluster NGC 6341 (M92)

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Abstract

Globular clusters are among the oldest objects in the universe. This report showcases photometric analysis of the globular cluster NGC 6341 (M92) using data taken in the Johnson B and V filters to find a lower limit on the age of the universe. The data was collected using the University of Birmingham Observatory's (UBO) telescope camera, SBIG STX-16803, which was then reduced using standard calibration frames, such as dark, bias, and flat frames, to account for various errors. Following this, source detection and aperture photometry processes were run on stacked science frames for the two bands using packages and functions from photutils. After gaining accurate fluxes at an aperture of 4.5 pixels, identical stars in both bands were calibrated to the apparent magnitude system using the standard star found in the data, HD 156964. The resulting colour-magnitude diagram reveals well-defined characteristics, such as the Red Giant Branch. Unfortunately, the data was unable to get to a low enough magnitude to display a clear turn-off point, despite that isochrones were fitted to the dereddened data to place a lower limit on the age of the universe as 11.5 ± 2 Gyr within reasonable errors.

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Introduction

Scientific Background

Star clusters are regions of space where stars that have formed from the same giant cloud of gas and dust are gravitationally bound together. For stars within the cluster, it is safe to assume that the stars born are all of the same age, distance, and chemical composition, which allows astronomers to isolate the effects of stellar evolution.^{[1][2]} Clusters are split into two categories: open and globular. The latter are much older, found in the galaxy's halo and, assumed to be involved with protogalaxies.^[3]

Globular clusters represent some of the earliest times of the formation of galaxies^[4], helping place lower limits on the age of the Universe.^[5] Not only do they offer insights for stellar evolution with their high density of stars but also aid in the research of the age of the Milky Way and the chemical composition of the galaxy's halo.^[4] By determining the age of the oldest cluster, astronomers can place a constraint on the epoch of the Big Bang.^[6]

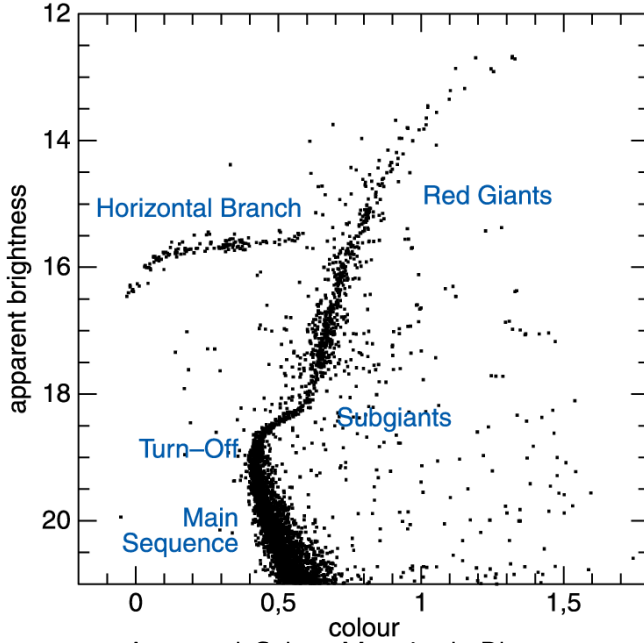


Figure 1. A general Colour-Magnitude Plot, indicating specific regions that show distinct patterns and behaviour.^[7]

A useful tool for the study of stellar populations is the Colour-Magnitude Diagram (CMD)^[8], which plots the apparent magnitude, m_V , against the colour index, as shown in Figure [1]. The colour index, $m_B - m_V$, shows the spectral type of the star. Typical values range from -0.3 for O-type stars, which are the hottest, whilst redder and cooler stars appear at a colour index of 2.^[9] This index, when plotted on a CMD, reveals the evolutionary tracks for the cluster.

Stars spend the majority of their lifetime on the main sequence. The timescale, τ_{MS} , for the main sequence is related to the mass, M , of the star by:

$$\tau_{MS} \propto M^{-2.5}$$

meaning that the more massive stars have less time on the main sequence branch^[10]; they are the first to evolve towards the red giant branch (RGB). Therefore, by identifying the Main Sequence Turn-Off (MSTO) point's magnitude and index, isochrone models can be fitted to find the best estimate for the age of the cluster.

To compare observational data to theoretical models, the raw CCD data must be turned into standard magnitudes. Once the data has been reduced of instrumental noise, the flux of the star, measured in ADU,

is converted to the instrumental magnitude using the relation:

$$m_{inst} = -2.5 \log_{10}(\text{Flux}) \quad (1)$$

However, instrumental magnitude is dependent on the telescope used, so it must be converted to apparent magnitude. This is achieved by observing a standard star of known magnitude within the image, m_{std} . The difference between the standard and instrumental is known as the zero point, ZP:

$$\text{ZP} = m_{std} - m_{inst} - k_\lambda X \quad (2)$$

where k_λ is the extinction coefficient for the respective wavelength and X is the airmass, which corrects for atmospheric extinction through Bouguer's Law.^[11]

Once obtaining the zero point for the standard star, all stars can be converted to the apparent magnitude by rearranging the same equation.

The majority of the CMDs have absolute magnitude rather than apparent magnitude, M , this is given by the distance modulus equation:

$$M = m_V - 5 \log_{10} d + 5 \quad (3)$$

where d is the distance in parsecs.

Project Aims

This project is centred on Messier 92 (NGC 6341), a metal-poor halo cluster located in the constellation Hercules. M92 is an cluster, being approximately 11.5 ± 1.5 Gyr old, and its low metallicity makes it a great tool for testing stellar evolution and age-dating.^[12] The primary aim of this work is to complete photometric analysis of data and obtain an estimate of the age by fitting model isochrones. To achieve this, the project has the following objectives:

1. Process raw CCD images in the B and V bands and correct them for instrumental signatures by using typical frame reductions.
2. Extract instrumental magnitude for identical cluster members in both bands, while accounting for background noise and source crowding.
3. Standardise the instrumental magnitudes to apparent by using a standard star to calculate the zero point.

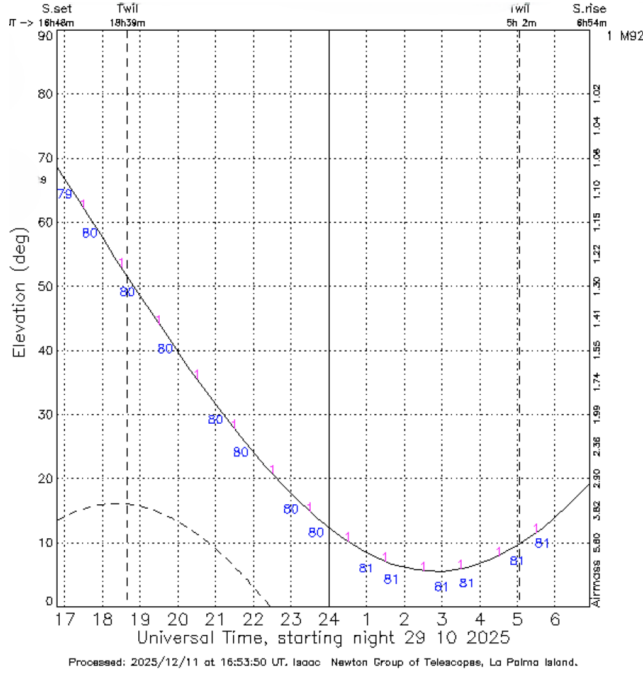


Figure 2. STARALT plot for Cluster M92 (NGC 6341) at the UBO for the night of 29th October 2025, showing the change in elevation and airmass throughout the night.

4. Ultimately, plot a CMD for the cluster members, and compare it to theoretical isochrone models to determine the age of the cluster.

1. Observational Data

1.1 Instrumentation

Data was collected from the University of Birmingham Ritchey-Chretien Alluna 20in Telescope using the SBIG STX-16803 camera, providing a field of view of approximately half a degree with a pixel scale of 0.467 arcseconds per pixel.^[13] Observations were carried out using standard Johnson-Cousins *B* (Blue) and *V* (Visible) band filters.^[14] These were selected as they are the most sensitive compared to the *U* and *I* filters for moderate aged stars. Furthermore, these filters are very common in observing clusters, so it was better to have direct comparisons to standard isochrones and CMD found in literature.

1.2 Target Selection

The subject M92 was chosen from a catalogue of globular clusters for its favourable observability during cycle 13, which took place from late October to late November 2025. M92 is located at Right

Ascension $\alpha = 17^h 17^m 07.39^s$ and a Declination of $\delta = +43^\circ 08' 09.4''$ ^[15], making it a great target given the latitude of the observatory is $+52^\circ 23' 14.8''$. From Figure [2], the STARALT plot^[16] for the projected elevation of M92 remains fairly above the horizon for the ideal observation time. In addition to this, during the observation time M92 remains at a sufficiently low air mass, $X < 2.0$, throughout. The airmass was imperative for completing accurate photometry to minimise atmospheric extinction of sources, especially for the dense core.

The primary challenge was to observe faint sources of magnitude $V > 18$ to ensure that the MSTO was observed while also ensuring a sufficient signal-to-noise ratio $\text{SNR} > 5$.

1.3 Data Acquisition

During the observation window of cycle 13, the first set of data was taken on the 29th of October; this was under grey time. Due to time constraints, only the *V*-band data was taken as well as the calibration frames for this day. The second set of observations in the *B*-band was taken on the 19th of November; this was during dark time.

To correct for instrumental signatures and inherent flaws in the telescope system, a number of calibration frames were taken on both nights of observation. The first set of frames taken were the bias frames; although the bias and dark frames are inherent to the CCD and are independent of the filter, they were taken on both nights to identify any outliers on the night of observation. Bias frames are zero-second exposures taken with the shutter closed to account for the read noise, the noise during the readout process. A large number of these frames were taken to minimise statistical noise. $N_{bias}^V = N_{bias}^B = 20$.

The second of the calibration frames taken were the dark frames; these are scaled exposures taken whilst the shutter is closed. These are essential to map the dark current, which is the thermally generated electrons that accumulate over time as a function of temperature. As a result, the number of dark frames collected was $N_{dark}^V = N_{dark}^B = 5$, and all of them had an integration time of 30 seconds to easily scale up to science frames.

Lastly, the final set of calibration frames were the flat field frames, which are dependent on the filters as well as the optical system used. These frames are taken of uniformly illuminated sources, which for the

observations was the sky during the twilight hour, and measure the sensitivity variations in pixels as well as imperfections in the optical system such as vignetting and dust halos. In the V -band, the number of frames taken was $N_{flat}^V = 11$ with each exposure being 5 seconds, whilst in the B -band, the number of frames taken was $N_{flat}^B = 5$ with exposure being 10 seconds.

The actual set of observations in both bands differed as well due to the moon phase; as a result, shorter observations were needed for the B -band dataset, as it was dark time, so less exposure time was needed to achieve the same SNR. However, to improve the SNR as well as utilise time at the observatory, more exposures than needed were taken for the B -band. The number of science frames taken in the V -band was $N_{science}^V = 30$, and each had an exposure time of $t_{exp} = 300$ seconds. On the other hand, the B -band data has $N_{science}^B = 35$ with $t_{exp} = 60$ seconds, but only 25 were needed to achieve the targeted SNR.

1.4 Data Reduction

The raw data was processed using the Python packages `ccdproc` and `astropy.nddata`.^[17] `CCDDData` to not only access the data but also compute statistical values such as means and standard deviations. Before beginning to conduct photometric analysis of the science frame, they had to be reduced by the calibration frames. To produce a reduced science frame, the general reduction procedure, given by Equation [4], where M denotes the master frame and SM denotes a scaled master frame, was used.

$$\text{Science}_{reduced} = \frac{\text{Science}_{raw} - \text{Bias}_M - \text{Dark}_{SM}}{\text{Flat}_M} \quad (4)$$

1.4.1 Bias Frame

Firstly, there was the creating of the master bias frame. Initially, the frames were checked for any outliers, and the mean and standard deviation for each frame were printed, but no outliers for either band were found. After the checks, the frames were median combined, as this statistic effectively removes large outliers. The master bias frame for the V -band gave a mean value of 947.34 ± 5.39 ADU per pixel, whilst the B -band returned a value of 941.41 ± 5.39 ADU per pixel; this is a fairly consistent result. This shows how, despite the same instrument being used, on different nights it yields different results.

1.4.2 Dark Frame

Following the bias frame, the quality same check was done for the dark frames, revealing no significant outliers. Before median combining, each individual dark frame was bias subtracted to isolate the thermal noise. The resulting master dark frame returned mean counts of 0.0491 ± 50 ADU for the V -band and 0.688 ± 42.423 ADU for the B -band.

These measured values reveal a significant discrepancy when compared to the manufacturer's specification. The camera specification states a value of $0.295e^-/\text{pixel}/\text{sec}$ at -5°C . Given that the observations were conducted when the CCD temperature was set to -9.96°C , and assuming that the temperature and dark current are exponentially related, the expected thermal noise should be $0.1475e^-/\text{pixel}/\text{sec}$. In contrast, the values obtained the dark current are too small with $I_V = 0.0023e^-/\text{pixel}/\text{sec}$ and $I_B = 0.0316e^-/\text{pixel}/\text{sec}$.

These extremely small values, combined with the standard deviation that exceeds the signal, suggests that the dark current can be considered negligible. Subtracting further frames by the master dark would only add additional noise, this should have been omitted but were still used to ensure the removal of hot pixels.

1.4.3 Flat Frame

The development of the master flat frame followed the same quality control similar to that of the bias and dark frames. Initial inspection of the raw flats in the B band showed no significant outliers, though a gradual trend in the mean count was observed. This variation is expected for twilight flats, as the sky brightness changes rapidly during the acquisition window. The individual frames remained consistent within 1σ , confirming their suitability.

However, this was inconsistent with original V -band flat dataset. The gradual trend was not observed and individual frames did not remain consistent to 1σ . After further inspection in DS9^[18], it was revealed that the frames were not taken during astronomical twilight. Figure [3], shows that the sky background has dimmed sufficiently to reveal point sources. Since flat frames represent uniformly lit surface, the presence of these stars would introduce holes in the final reduced images. To rectify this, this data set was not used and replaced with an alternative set of flats taken correctly during twilight on a different night.

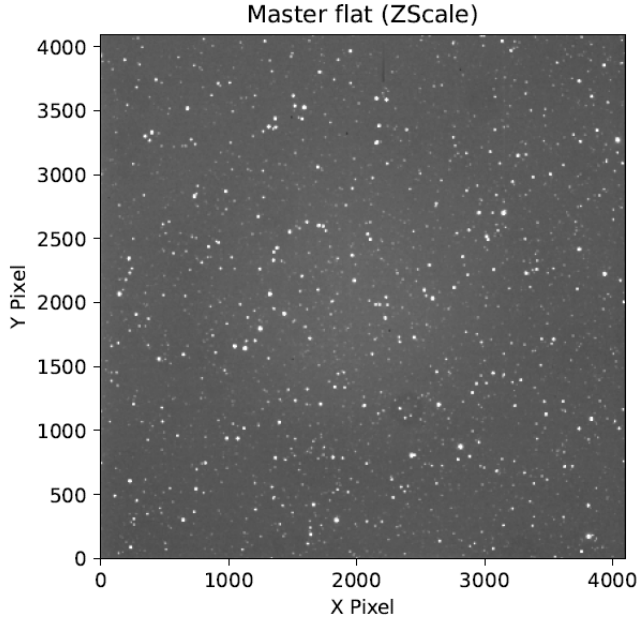


Figure 3. Master Flat frame for flat frames taken in the *V*-band on the night of observations. The image reveals point sources embedded in the flat field data, as a result these set of flats were removed.

For the final processing, all flat frames were bias and dark subtracted to remove additional noise. To preserve the flux scale, each frame was normalised to unity by dividing by their respective mean. These normalised frames were then median combined to create the master frame for each band.

1.4.4 Science Frame

Ultimately, Equation [4] was used to reduce all the science frames. However, this took a bit longer, as each frame was checked individually on DS9 for any outliers such as cosmic rays, cloud coverage and other transient events. The same check as before was done, as it could quickly identify large disruptions such as clouds, but for smaller, harder transients across the cluster, individual checks were necessary. As a result, 5 frames were removed from the *V*-band dataset; 4 frames were removed due to spontaneous clouds over the cluster. An additional frame was removed after careful analysis, as there was a cosmic ray directly over the cluster. Although median combining frames would remove outliers like this, extra precaution was taken for this frame as it was directly over the cluster; hence, this frame was removed. There were no outliers for the second dataset.

Using the reduced science frames, these were also

stacked together to further improve the SNR and detect fainter stars. However, initial attempts to stack were proven unsuccessful, as the small shifts in the telescope tracking were not taken into account. This was more dominant in the *V*-band as the exposure time was much larger than the *B*-band; hence, the shift is bigger between frames. To counteract this, the *astrowarp*^[19] library was used; this took in manual inputs, which were the three brightest stars in each frame, and compared them to a reference frame and aligned each frame accordingly. This was done to both datasets as shown by Figure [4], even though the *B*-band didn't need it as much.

Now that both master science frames were obtained, photometric analysis could technically be conducted. However, before further analysis, both science frames would have to be spatially aligned as well. Since the data sets were taken on different days, the alignment of the telescope differs as well; hence, this would have to be corrected. This ensures that star fields overlay perfectly, making it significantly easier to identify and cross-reference identical stars when plotting colour-magnitude diagrams. As shown in Figure [4c], the alignment of the *B*-band was significant to the upper-right; this process is also vital as fluxes are not conserved whilst aligning.

2. Analysis

2.1 Source Detection

The next step for each band was to identify the sources on each diagram. Since the *V*-band was taken under grey time and had a lower amount of frames taken to reach the required SNR, the number of sources that would have been able to be resolved was going to be lower than the *B*-band.

Photometry was performed using the *photutils* package.^[20] Due to the high stellar density of the globular cluster, a simple global background subtraction was insufficient. Instead, a 2D background model was constructed using a box size of 512×512 , and sigma clipping at 3σ ; this created a more accurate local sky background.

From Figure [12a], the background shows a clear gradient from the bottom right to the upper left of increasing intensity; this could be an effect of the moonlight as well as vignetting. Whilst Figure [12b] does not show a clear gradient and is rather almost uniform, there is a clear corner of lower intensity, possibly

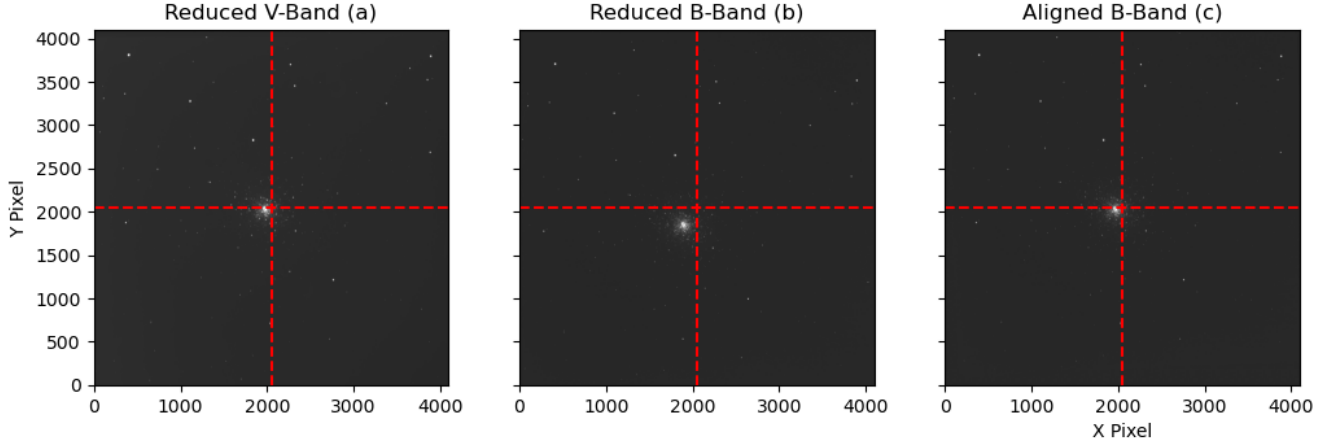


Figure 4. [a]: The final stacked master science frame in the *V*-band. [b]: The final stacked master science frame in the *B*-band. [c]: The aligned master *B* science frame to the co-ordinates of *V*-band's stars.

due to the vignetting or gradient in the sky brightness. The background of the *V*-band gave a median of 1683.81 counts with an RMS of 23.47, whilst the *B*-band gave a value of 125.12 with an RMS of 2.77. Although the difference in values is quite drastic, the difference is normal given that the *V*-band data was taken during grey time whilst the *B*-band was during dark time; therefore, the moon, although not full, was raising the background level significantly.

Sources were identified from the background-reduced frames by creating a segment map for sources detected over a threshold. Since the *B*-band data was taken over a dark time and has a lower background median, the threshold set was only 3σ , where σ represents the background RMS; this ensured that the detect source function still avoided noise in the background but picked up fainter sources. The function was also told to detect sources with at least 3 other pixels; this allowed for fainter and smaller sources nearer to the core to be detected. The sources were then deblended, which was crucial for sources nearer to the core, as many stars would be grouped together and appear as larger sources than they actually were. The segmentation image as shown in Figure [13a] indicates how the core of the cluster transforms from one large source to smaller point sources shown in Figure [13b]; the number of sources the segmentation image reveals is 1300, and after deblending, this value increases to 1826.

Similarly, the process was completed for the *V*-band data, but for this case the threshold was set higher at 5σ as the background was much higher.

This still resulted in noise on the image but was a limit set to avoid losing faint stars nearest to the cluster. As mentioned before, due to removing frames and the time of observations, the number of sources detected was lower for this band. Segmentation only retrieved 1010 sources, whilst the deblending only increased it to 1285 sources.

A source table was created for both bands, indicating the pixel position on the CCD, the segment's flux, the error on the flux and the area of the source in pixels squared. The error was calculated using the effective gain on the CCD detector, which was found on the header of the frames as 1.38. Following this, the next step was to locate identical sources, as when plotting CMD diagrams, the colour index must be the subtraction of the magnitude of the same source in the two bands.

For this reason, an analysis on an annulus region was necessary; this was due to the core being heavily dense and proving to be far too difficult to pick out identical sources. A core radius of 215 pixels was chosen based on observations in both bands for the number of sources that were blended together. For the outer cluster radius, a value of 775 pixels was chosen; this value was chosen to avoid stars that were known to not be in the cluster. As shown in Figure [14], the left column displays this annulus region, as well as the star that was avoided located at roughly $(x,y) = (1850,2800)$. This annulus region includes all the stars within the cluster but also has stars in the background; unfortunately, these stars were still considered due to time constraints.

Following this, both catalogues were compared and stars within the annulus region were picked out. This was done using cKDTree function from SciPy^[21], which ensured there was only one match between each catalogue within a 1.5 pixel limit since both frames were aligned. In total, both frames had 535 common stars, with 287 common within the annulus. These stars, as shown in Figure [14], were identical. The function did miss a few stars in the region which are detectable by eye, this was likely due to the pixel coordinates being more than a 1.5 pixels away from each other, indicating the alignment was not perfect for all sources.

2.2 Flux Calibration

Before converting the sources' flux to magnitudes, they had to be calibrated to apparent magnitudes by finding the zero point in each filter. This was done by identifying a known star within the frame; from the analysis, the easiest source to identify was located through the diagonal pattern of stars in the top left in Figure 4a. Using Ref.[22], the upper left star was identified as HD 156964 (HIP 84658), with an apparent magnitude of $m_V = 8.655 \pm 0.003$ and $m_B = 9.745 \pm 0.003$.^{[23][24]}

2.2.1 V-Band

Aperture photometry was conducted on the star to determine a more accurate flux than the segment. This was done as the given flux does not consider the whole brightness of the star. Aperture photometry was iterated over different radii to obtain a growth curve^[25], as shown in Figure [5] by the dotted line. This curve measures the flux at different radii to illustrate at what pixel size to stop measuring the star's brightness without background being included.

However, this curve did not flatline and continued to rise at increasing radii, indicating that the background was still noisy. To correct for this, aperture photometry was conducted at a large radius to obtain a median residual sky background, S_{sky} ; this brought a value of 84.86 ± 0.52 ADU per pixel. The corrected flux, F_{corr} , was then obtained using the following equation:

$$F_{corr} = F_{ap} - (S_{sky} \times \text{Area}) \quad (5)$$

where F_{ap} is the flux measured within the aperture. As shown in Figure [5], the corrected graph for this residual background better shows the flatline effect when measuring the stars' flux at larger radii.

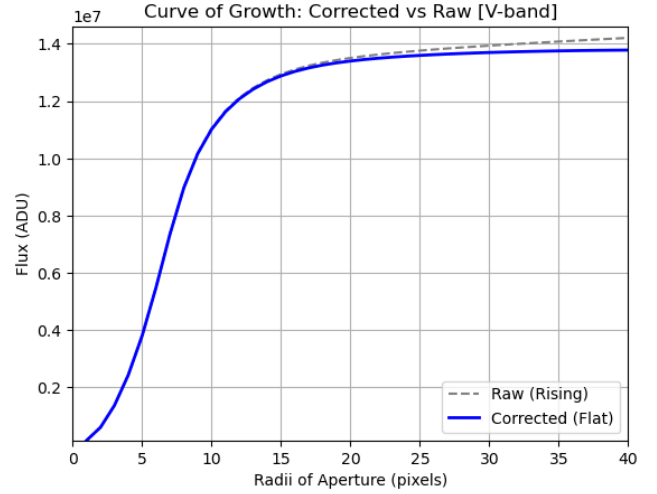


Figure 5. Growth Curve of V-Band of Flux(ADU) measured across different radii (pixels) using aperture photometry. The dotted line represents the normal fit data that had been 2D background subtracted whilst the solid blue line represents the residual sky background subtraction data

To gain an accurate radius at which the background is not measured while encompassing the entirety of the star's flux, annulus photometry was used. This process was iterated over a small number of radii to return the pixel mean pixel value at each annulus radius. This value was compared to the original value of the background median of 1683; when the mean of the annulus was less than the background median, it was enough to assume that the entirety of the star's flux was measured. In the case for the V-band, as shown in Figure [6], an annulus of radius $R_{in} = 15$ and $R_{out} = 16$ (shown by the purple and red circle) has a brightness of 1716, whilst an annulus with radius $R_{in} = 16$ and $R_{out} = 17$ (shown by the purple and blue circle) has a mean value of 1275. Therefore, a precise radius of 16 pixels accurately measures the star's flux.

Using the aperture flux at a radius of 16 pixels and converting to the instrumental magnitude given by Equation [1], the instrumental magnitude was subtracted from the apparent magnitude using the value found in Ref.[24], with an extinction coefficient of $k_V = 0.1$ mags/airmass^[13] and average airmass from the science frames of $X = 1.589$, using Equation [2] An accurate zero point was calculated for the V-band as $ZP_V = 26.204 \pm 0.003$.

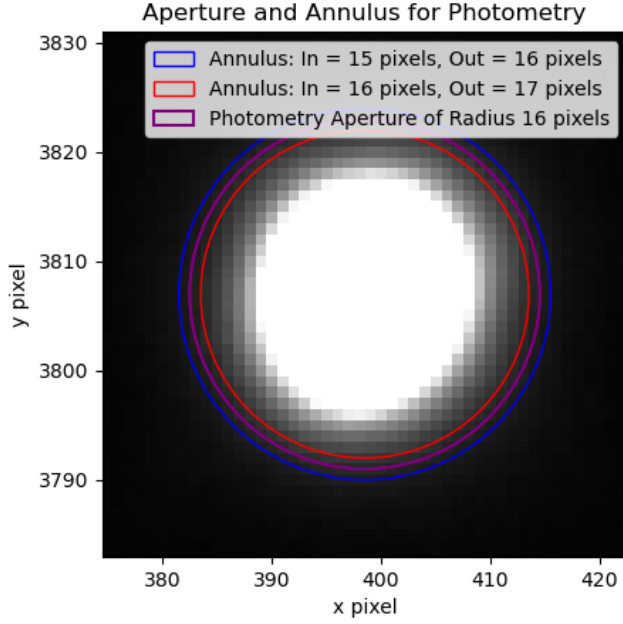


Figure 6. Annulus and Aperture Photometry for V-Band data for HD 156964 at different radii. Both annulus meet at $r = 16$ pixels hence the purple colour to show the overlap.

2.2.2 B-Band

The procedure was applied to the *B*-band data; however, as discussed prior, the background of this data is not as noisy as the *V*-band. For this case, the initial growth curve was showing the flatline at larger radii as shown by Figure [7], which indicated that the correction was not needed. The correction was still performed to confirm this and returned a residual background median of 1.08 ADU per pixel. For accuracy of results, it was still performed for the rest of the calculation for the zero point. To gain the accurate radii, the same annulus photometry sequence was carried out; this time an annulus of radius $R_{in} = 11$ and $R_{out} = 12$ (shown by the purple and red circle) has a brightness of 171, whilst an annulus with radius $R_{in} = 12$ and $R_{out} = 13$ (shown by the purple and blue circle) has a mean value of 116, so when comparing this to the original median of the background of 125, a suitable radius of 12 was chosen.

Following this Equation [1] and [2] were used, this time with values of $k_B = 0.2$ mags/airmass^[13] and average airmass of $X = 1.355$, to return a value for the zero point as $ZP_B = 24.650 \pm 0.003$.

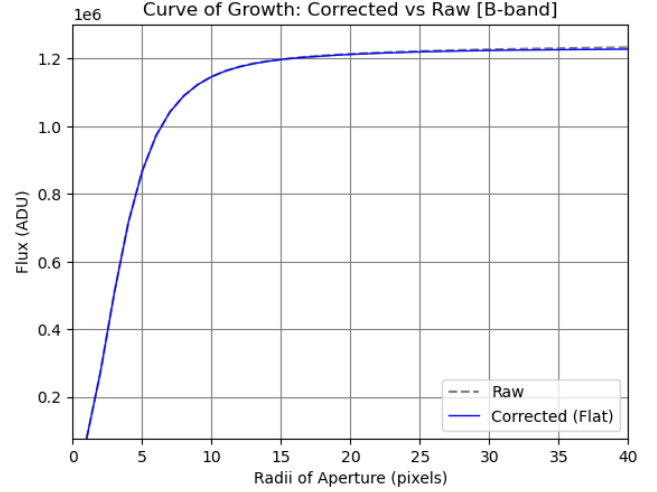


Figure 7. Growth Curve of *B*-Band of Flux(ADU) measured across different radii (pixels) using aperture photometry. The dotted line represents the normal fit data that had been 2D background subtracted whilst the solid blue line represents the residual sky background subtraction data, but there is minimal change overall.

3. Results

After obtaining the zero point in both bands, the segment fluxes of common stars were converted to apparent magnitudes. Then the colour index of each star was calculated, and this was plotted against the apparent magnitude in the *V*-band, ultimately giving the sought-after CMD plot as shown in Figure 8. This plot shows a clear band for the main sequence as well as the start of the horizontal branch with reference to Figure [1]. However, upon further analysis there were a number of errors within this plot:

1. The colour index was too low, as shown in table , typical colour indexes range from $-0.35 < C.I. < 2$, therefore a lot of stars were incorrectly shown on this diagram.
2. Data was not corrected for reddening; the flux that was measured was diminished by the inter-stellar medium.
3. The main sequence branch is not nearly steep enough when comparing it to the model CMD (Figure [1]), therefore it would unlikely fit any isochrone models to compare the data.

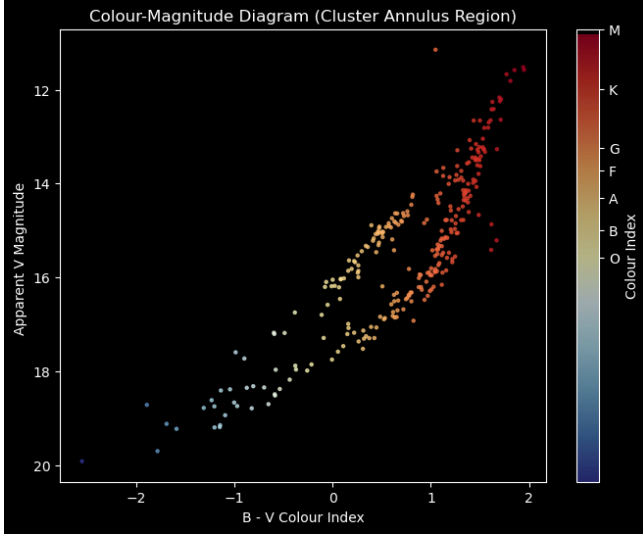


Figure 8. First plot of a Colour-Magnitude Diagram, using the apparent magnitudes from the segment flux of each star

3.1 Segment Flux Correction

After further inspection, it was wrong to use the segment flux of each star in separate bands to calculate the flux and magnitude. Segment flux measures the flux of the star within a certain amount of pixels, this segment is not constant for the same star in different bands nor does it encompass the entirety of the stars flux. Therefore to correct this, a fixed circular aperture photometry was used to ensure all sources in both bands were treated the same. The fixed radius of the aperture was chosen from seeing conditions and taken as $1.5 \times FWHM$ (Full Width at Half Maximum), giving a value of 4.5 pixels. This ensured that majority of the source's flux was captured while also ensuring minimal background was included.

While obtaining the flux at fixed radius aperture photometry, the fluxes were also corrected for the residual background that was done to the calibration star, as this was found to strongly affect the V -band data. A similar method was applied to all the sources, where an annulus at a larger distance than fixed radius returned a local sky residual median per pixel estimate. The total corrected flux for each source was derived by using Equation [5].

By applying a consistent aperture size to both the B and V datasets, the systematic errors affecting the shape of the CMD were corrected, as shown in Figure [9]. This adjustment significantly improved the photometric accuracy and resulted in a tighter and

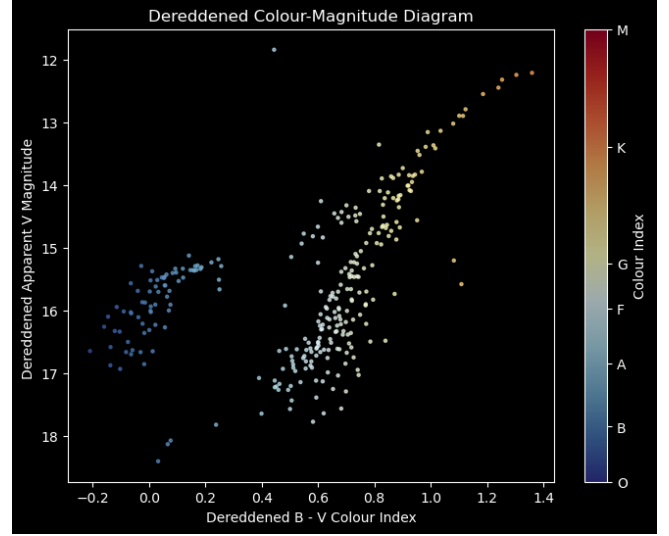


Figure 9. CMD of data after considering aperture photometry at a fixed radius for all sources in both bands, as well as correcting for interstellar extinction.

well defined Main Sequence and Horizontal Branch.

3.2 Reddening Correction

The observed magnitudes and colours derived previously were not corrected for interstellar extinction. Interstellar extinction or better known as reddening is caused by the scattering and absorption of light by dust grains within the galactic plane.^[26] This effect is non-uniform and affects blue light more than red. For this reason, M92 appears to be dimmer and redder than its intrinsic state. Therefore to compare this data to theoretical models, it was necessary to perform a correction for this.

$$E(B - V) = (B - V)_o - (B - V)_i \quad (6)$$

Reddening is quantified by the colour excess, E_{B-V} which is essentially the difference between the observed $(B - V)_o$ and intrinsic colour index $(B - V)_i$ as shown by Equation [6]. For the M92, this value is 0.025 ± 0.010 .^[12] Therefore to correct the colour indices on the CMD, Equation [6] is rearranged for the intrinsic colour index. To convert just the V -band magnitudes, the ratio of total to selective extinction at V is used^[27]:

$$A_V = R_V \times E(B - V) \quad (7)$$

Where R_V is the total extinction and A_V is the visual magnitude extinction, for objects located within

the halo of the Milky Way the total extinction tends to be around $R_V = 3.1$.^[28] Therefore using the values the visual extinction was calculated as $A_V = 0.062$ mag. Therefore to correct the V magnitudes a simple subtraction was done, $V_i = V_o - A_V$. This did not make a noticeable difference but did shift the data points to the left a small amount.

3.3 Isochrone Fitting

To derive the age of the cluster, theoretical isochrone models made from PARSEC^[29] found in Ref.^[30] were fitted to the observational data. The models selected were at a fixed metallicity of $Z = 0.0001$ or $[\text{Fe}/\text{H}] = -2.3$ dex which is consistent with the known chemical composition of this metal-poor cluster.^[12] The isochrones were fitted to the dereddened CMD by also taking into account colour excess of $E_{B-V} = 0.025$, and magnitude extinction of $A_V = 0.062$ mag. The isochrone was then shifted vertically by the true distance modulus, which is obtained when rearranging Equation [3] for $m_V - M$. For NGC 6341, this is approximated by using a distance of $d = 8.231 \pm 0.347$ kpc giving a value of $m_V - M = 14.6 \pm 0.1$.^[31]

Modelling isochrones for different ages, Figure [10], proved that the data matched significantly. From this, it can be seen that a number of ages fit the data, within approximation, due to time constraints a least squared function was not fitted to give a value for the age, so the average age of 11.5 ± 2 Gyr was determined from the isochrone fitting, as the orange band was the median of the ages towards the MSTO point.

4. Discussion

4.1 Analysis of Limitations

The most significant limitation of this study was the photometric depth of the dataset, specifically the V -band. During the calibration on the science frames, it was mentioned that 5 frames were removed due to transient events, this meant that the magnitude at the required SNR was not achieved. A further analysis of the SNR as a function of apparent magnitude, Figure [11], reveals that the 5σ limit of observations occurs at $V \approx 17.422$.

This is a physical challenge when trying to estimate the age of the cluster, because the MSTO typically occurs at $V \approx 18.5$ for M92.^[12] Consequently, this means that the data is limited before the required phase. This implies that the the isochrone fit is constrained to the red giant branch, rather than the end point of the

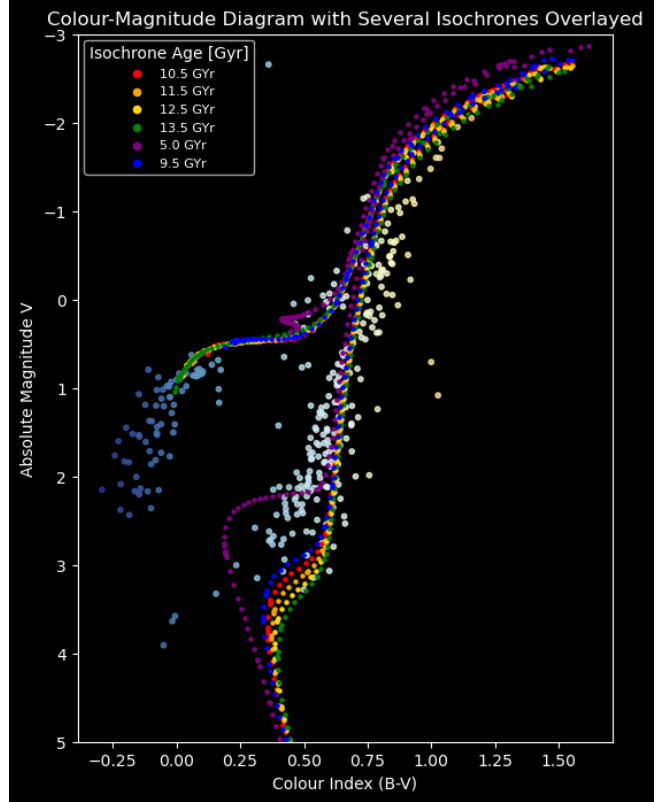


Figure 10. Isochrone fitting on the observed data for a range of ages. An age of 5 Gyr was used to demonstrate that young ages for the cluster is unsuitable as the RGB does not match. Ages between $10.5 < \tau < 13.5$ provide a good fit to the general shape.

main sequence. Since the RGB is less sensitive to age than the MSTO, this introduced a larger systematic uncertainty in the derived age.

Due to the location of UBO, being near to the city, the data suffers from atmospheric scattering and light pollution. Calibration stars within the same image as the data typically do not need the extinction term in Equation [2], since all stars are subject to the same column of air the ZP term accounts for this. However, without this specific adjustment, the derived magnitudes would not have aligned with the isochrone models, suggesting that the calibration of data was diminishing the magnitude of the stars in some form.

4.2 Future Ambitions

Future analysis of M92 should replace aperture photometry with Point Spread Function (PSF) fitting. Making use of packages like DAOPHOT^[32] or libraries in astropy^[17] to deblend overlapping stars would sig-

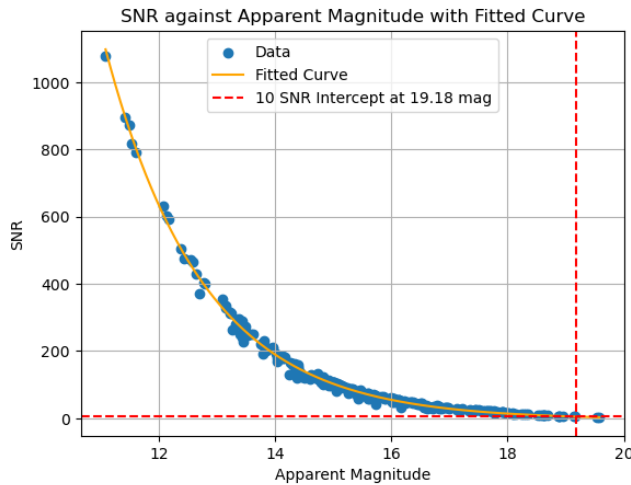


Figure 11. SNR against Flux measured for the V-band data. An exponential decay was fitted to the data points to obtain the interception of the minimum 5σ SNR, this output a value of 17.41 mag. Clearly showing that V-band data did not go deep enough while maintaining good SNR.

nificantly improve accuracy for the crowded core. Additionally, acquiring a surplus of science frames would have safeguarded the target SNR.

5. Conclusion

This study successfully characterised the stellar population of the globular cluster NGC 6341 (M92) using calibrated ground-based photometry in the B and V filters. The construction of a precise colour–magnitude diagram revealed the cluster's Red Giant Branch and Horizontal Branch, allowing for comparison with theoretical isochrone models (Figure [10]). Although the Main Sequence Turn-Off was not observed, fitting the metallicity and distance parameters found in literature produced an estimated cluster age of 11.5 ± 2 Gyr, consistent with published values of 11.5 ± 0.5 Gyr.

In summary, these results show how ground-based photometry can produce results of stellar evolution in clusters that are accurate when carefully calibrated and adjusted for systematic errors. This test effectively demonstrates how stellar systems change over time and provides a baseline for the age and makeup of the Milky Way's halo by comparing theoretical models with observational data. The study sets the foundation for future photometric analysis aimed to improve cluster member analyses and tests the sensitivity and limitations of ground-based observations.

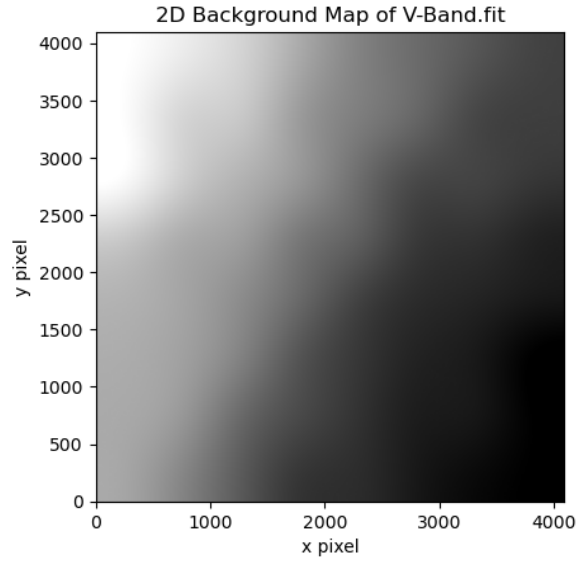
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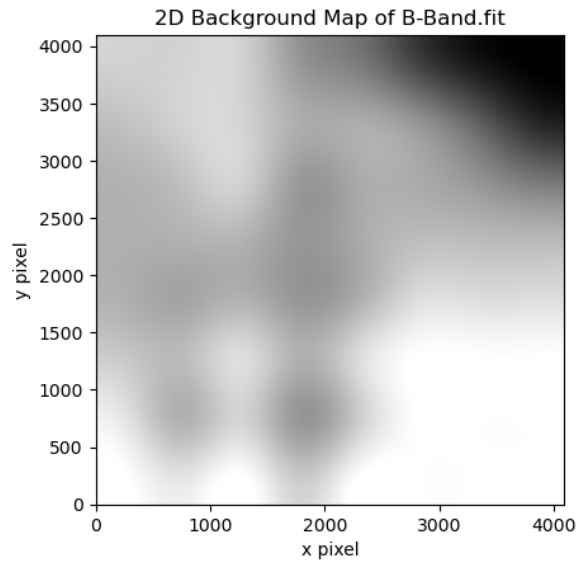
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Appendix

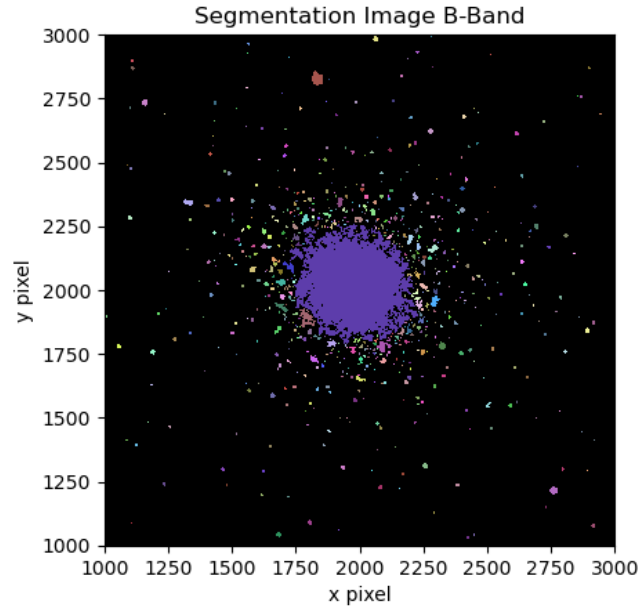


(a) 2D background for [V-band]

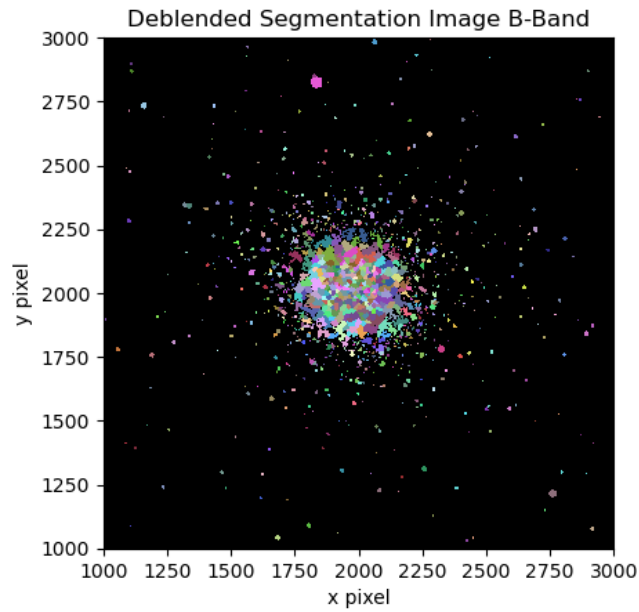


(b) 2D background for [B-band]

Figure 12. 2D Background image created for both bands of data. This was done using a box size of 512, filter size of 5, sigma clipping to 3σ and estimating the background using the median.



(a) Segmentation Map of all the sources identified in the *B*-band



(b) Deblended Segmentation map for sources that are close together.

Figure 13. A comparison for how strongly a deblended segmentation map was needed when trying to identify sources with the cluster

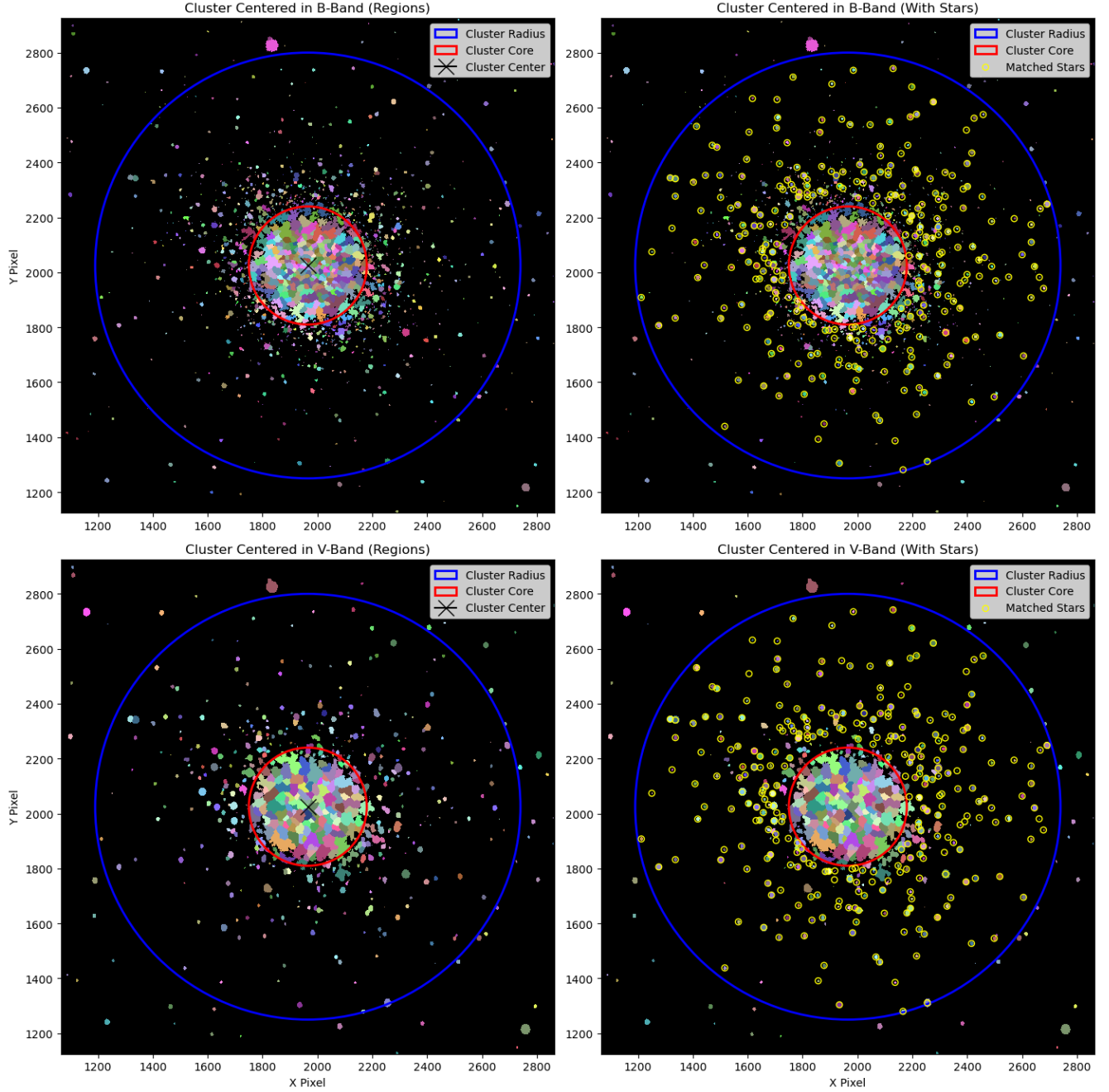


Figure 14. [Left Column]: Annulus region for the cluster, with a core radius of 225 pixels and an outer tidal radius of 775 pixels. Avoiding the bright pink segment at the top of both images [Right Column]: Annulus region for cluster with the identified 287 matched sources.